# QuickPhone — Physical ERD (complete): 16 tables and all keys and attributes

One diagram for the whole data model. Each verb reads from the "one" side to the "many" side. PK primary key · FK foreign key (→ table) · UK unique · CK checked value list · NN not null · derived = denormalized

**store** — D7 Staff & Stores

| | | | |
|---|---|---|---|
| PK | **store_id** | INTEGER | NN |
| UK | store_name | VARCHAR(60) | NN |
| | address | VARCHAR(200) | NN |
| | phone | VARCHAR(20) | NN |
| | email_address | VARCHAR(120) | NN |
| | business_hours | VARCHAR(120) | |

**employee** — D7 Staff & Stores

| | | | |
|---|---|---|---|
| PK | **employee_id** | INTEGER | NN |
| FK | store_id | INTEGER | → store |
| | first_name | VARCHAR(40) | NN |
| | last_name | VARCHAR(40) | NN |
| UK | email | VARCHAR(120) | NN |
| CK | role | VARCHAR(12) | NN |
| | password_hash | VARCHAR(255) | NN |
| | active_flag | BOOLEAN | NN |

**supplier** — D6 Suppliers

| | | | |
|---|---|---|---|
| PK | **supplier_id** | INTEGER | NN |
| UK | supplier_name | VARCHAR(100) | NN |
| | contact_name | VARCHAR(80) | |
| | phone | VARCHAR(20) | |
| | email | VARCHAR(120) | |
| | active_flag | BOOLEAN | NN |

**purchase_order** — D5 Purchase Orders

| | | | |
|---|---|---|---|
| PK | **purchase_order_id** | INTEGER | NN |
| FK | store_id | INTEGER | → store |
| FK | supplier_id | INTEGER | → supplier |
| FK | created_by_id | INTEGER | → employee |
| FK | approved_by_id | INTEGER | → employee |
| CK | po_status | VARCHAR(20) | NN |
| | created_at | TIMESTAMP | NN |
| | approved_at | TIMESTAMP | |
| | placed_at | TIMESTAMP | |
| | rejection_reason | VARCHAR(300) | |

**appointment** — D2 Appointments

| | | | |
|---|---|---|---|
| PK | **appointment_id** | INTEGER | NN |
| FK | customer_id | INTEGER | → customer |
| FK | store_id | INTEGER | → store |
| FK | technician_id | INTEGER | → employee |
| | scheduled_at | TIMESTAMP | NN |
| CK | appt_status | VARCHAR(12) | NN |

**customer** — D1 Customers & Devices

| | | | |
|---|---|---|---|
| PK | **customer_id** | INTEGER | NN |
| | first_name | VARCHAR(40) | NN |
| | last_name | VARCHAR(40) | NN |
| IX | phone | VARCHAR(20) | NN |
| | email | VARCHAR(120) | |
| | created_at | TIMESTAMP | NN |

**device** — D1 Customers & Devices

| | | | |
|---|---|---|---|
| PK | **device_id** | INTEGER | NN |
| FK | customer_id | INTEGER | → customer |
| | device_type | VARCHAR(20) | NN |
| | brand | VARCHAR(40) | NN |
| | model | VARCHAR(60) | NN |
| IX | serial_or_imei | VARCHAR(40) | |

**repair_job** — D3 Repair Jobs

| | | | |
|---|---|---|---|
| PK | **repair_job_id** | INTEGER | NN |
| UK | ticket_number | VARCHAR(12) | NN |
| FK | device_id | INTEGER | → device |
| FK | store_id | INTEGER | → store |
| FK,UK | appointment_id | INTEGER | → appointment |
| FK | assigned_tech_id | INTEGER | → employee |
| CK | status | VARCHAR(20) | derived · NN |
| | reported_issue | VARCHAR(500) | NN |
| | intake_condition | VARCHAR(500) | |
| | diagnosis_notes | VARCHAR(2000) | |
| | repair_notes | VARCHAR(2000) | |
| | check_in_at | TIMESTAMP | NN |
| | completed_at | TIMESTAMP | |
| | final_cost | DECIMAL(10,2) | |
| | email_sent_at | TIMESTAMP | |
| FK | email_sent_by_id | INTEGER | → employee |
| CK | email_status | VARCHAR(10) | |
| | picked_up_at | TIMESTAMP | |
| | id_verified_flag | BOOLEAN | NN |
| | signature_file | VARCHAR(255) | |

**job_status_history** — D3 Repair Jobs

| | | | |
|---|---|---|---|
| PK | **history_id** | BIGINT | NN |
| FK | repair_job_id | INTEGER | → repair_job |
| CK | status | VARCHAR(20) | NN |
| FK | changed_by_id | INTEGER | → employee |
| | changed_at | TIMESTAMP | NN |
| | note | VARCHAR(300) | |

**quote** — D3 Repair Jobs

| | | | |
|---|---|---|---|
| PK | **quote_id** | INTEGER | NN |
| FK | repair_job_id | INTEGER | → repair_job |
| | labor_amount | DECIMAL(10,2) | NN |
| | parts_amount | DECIMAL(10,2) | NN |
| | total_amount | DECIMAL(10,2) | derived · NN |
| CK | decision | VARCHAR(10) | NN |
| | decision_at | TIMESTAMP | |
| FK | decision_by_id | INTEGER | → employee |

**po_line** — D5 Purchase Orders

| | | | |
|---|---|---|---|
| PK | **po_line_id** | INTEGER | NN |
| FK | purchase_order_id | INTEGER | → purchase_order |
| FK | part_id | INTEGER | → part |
| | quantity_ordered | INTEGER | NN |
| | quantity_received | INTEGER | NN |
| | unit_cost | DECIMAL(10,2) | NN |

**payment** — D3 Repair Jobs

| | | | |
|---|---|---|---|
| PK | **payment_id** | INTEGER | NN |
| FK | repair_job_id | INTEGER | → repair_job |
| | amount | DECIMAL(10,2) | NN |
| CK | method | VARCHAR(12) | NN |
| CK | payment_status | VARCHAR(10) | NN |
| | paid_at | TIMESTAMP | NN |

**job_part** — D3 Repair Jobs

| | | | |
|---|---|---|---|
| PK | **job_part_id** | INTEGER | NN |
| FK | repair_job_id | INTEGER | → repair_job |
| FK | part_id | INTEGER | → part |
| FK | purchase_order_id | INTEGER | → purchase_order |
| | quantity | INTEGER | NN |
| CK | part_status | VARCHAR(8) | NN |
| | recorded_at | TIMESTAMP | NN |

**part** — D4 Parts Inventory

| | | | |
|---|---|---|---|
| PK | **part_id** | INTEGER | NN |
| FK | default_supplier_id | INTEGER | → supplier |
| | part_name | VARCHAR(100) | NN |
| | device_model | VARCHAR(60) | |
| | grade | VARCHAR(20) | |
| | standard_cost | DECIMAL(10,2) | NN |
| | active_flag | BOOLEAN | NN |

**store_part_stock** — D4 Parts Inventory

| | | | |
|---|---|---|---|
| PK,FK | **store_id** | INTEGER | → store |
| PK,FK | **part_id** | INTEGER | → part |
| CK | quantity_on_hand | INTEGER | derived · NN |
| | reorder_point | INTEGER | NN |

**inventory_txn** — D4 Parts Inventory

| | | | |
|---|---|---|---|
| PK | **inventory_txn_id** | BIGINT | NN |
| FK | store_id | INTEGER | → store_part_stock |
| FK | part_id | INTEGER | → store_part_stock |
| FK | recorded_by_id | INTEGER | → employee |
| CK | txn_type | VARCHAR(12) | NN |
| | quantity_change | INTEGER | NN |
| | reason | VARCHAR(200) | |
| | txn_at | TIMESTAMP | NN |

schedules · employs · handles · books · leads to · receives · uses · owns · undergoes · generates · tracks · is used in · provides · fulfills · contains · appears in · appears on · logs

Not drawn, to keep the lines clear (each is an FK column in its table): the assigned technician of a REPAIR_JOB, JOB_PART.PurchaseOrderID, and the audit links in which an EMPLOYEE changes a status, decides a QUOTE, sends the completion email, records an INVENTORY_TXN, or creates or approves a PURCHASE_ORDER.